Fault Detection and Classification Based on Co-training of Semisupervised Machine Learning

Tamer S. Abdelgayed, *Student Member, IEEE*, Walid G. Morsi, *Senior Member, IEEE*, and Tarlochan S. Sidhu, *Fellow, IEEE*

***Abstract*—This paper presents a semisupervised machine learning approach based on co-training of two classifiers for fault classification in both the transmission and the distribution systems with consideration of microgrids. Unlike previous work in which only labeled data are treated using supervised machine learning approaches, this study uses a semisupervised machine learning approach to handle both labeled and unlabeled data. In order to extract the hidden features in the current and voltage waveforms, the discrete wavelet transform is applied, while the harmony search algorithm is utilized to identify the optimal parameters of the wavelets. The performance of the proposed method was examined on both transmission and distribution test systems in a simulation environment, and also using experimental hardware. The results have shown that the proposed approach provides flexibility and adaptability in dealing with various system conditions/configurations with high accuracy. The results also have demonstrated that the proposed semisupervised approach can improve the fault classification accuracy compared to that obtained using other machine learning approaches (i.e., supervised and unsupervised) in the case of utilizing unlabeled data to build and train the classifier's model.**

***Index Terms*—Fault diagnosis, fault protection, pattern recognition, semisupervised learning, wavelet transforms.**

Nomenclature

EMTP	Electromagnetic transients program.
FFT	Fast Fourier transform.
PSCAD	Power system computer-aided design.
EMTDC	Electromagnetic transients including dc.
Ca_j, Cd_j	Approximation and detail coefficients.
C_z	Closest neighbors label.
d	Sample difference.
f_{l0}, f_{l1}	Low- and high-pass filters.
g_{aj}, g_{dj}	Energy of approximations and details.
$p(\alpha\|\beta)$	Cases belonging to class α at a node β.
$P_{c1}(\delta\|N_r)$	Class probability estimation.
ph	Phases (A, B, and C).
S	Signal type (voltage or current).
Λ_m	Class label of the mth neighbors.
δ_u	Class label of the uth records in L_e.
E	Number of records in a leaf node L_e.
ε_t	Total number of classes.
τ_n	New record label.
Ω_v	Class labels set.

Manuscript received January 27, 2017; revised April 4, 2017 and May 18, 2017; accepted June 11, 2017. Date of publication July 13, 2017; date of current version December 8, 2017. *(Corresponding author: Walid G. Morsi.)*

T. S. Abdelgayed and T. S. Sidhu are with the Faculty of Engineering and Applied Science, University of Ontario Institute of Technology, Oshawa, ON L1H 7K4, Canada (e-mail: Tamer.SayedAbdelhamid@uoit.ca; tarlochan.sidhu@uoit.ca).

W. G. Morsi is with the Department of Electrical, Computer, and Software Engineering, University of Ontario Institute of Technology, Oshawa, ON L1H 7K4, Canada (e-mail: walidmorsi.ibrahim@uoit.ca).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TIE.2017.2726961

I. Introduction

SUPERVISED machine learning (SML) has been widely used for fault classification in electric power systems. In SML approaches, all the data must be labeled. According to [1], a large number of power system events are being recorded without labels, and the presence of such unlabeled data would be problematic to the SML approaches as they can only handle labeled data, which will lead to fault misclassification. The work presented in this study introduces a semisupervised machine learning (SSML) method that uses co-training of decision tree (DT) as an eager learner and k-nearest neighbor (KNN) as a lazy learner to handle the presence of unlabeled data and applied to fault classification in electric power systems.

A. Previous Work

The previous work addressing the problem of fault classification in electric power system may be categorized into non-machine-learning-based approaches [2]–[4] and machine-learning-based approaches [5]–[17].

1) Non-machine-Learning-Based Approaches: Rahmati and Adhami [2] used a criterion index to classify the double phase to ground and the single phase to ground faults. The study relied on the values of zero and negative sequences of the current and voltage waveforms. In [3], a combination of the principle component analysis and the sequence component analysis was used to classify and locate the faults. In [4], the features in the current signals are extracted using the multiresolution analysis (MRA) and are used for fault classification. It is worth noting that the studies [2]–[4] relied on an arbitrary thresholds and these thresholds were mainly preset based on a trial and error process and without any justification.

2) Machine-Learning-Based Approaches: Studies [5] and [6] used the support vector machine (SVM) method

0278-0046 © 2017 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

Authorized licensed use limited to: Gheorghe Asachi Technical University of Ia¿i. Downloaded on January 26,2026 at 13:27:32 UTC from IEEE Xplore. Restrictions apply.

for fault classification. The approach presented in [5] relied on a combination of FFT and SVM, whereas FFT was used to extract the signals' features, which are then used as an input to SVM. Livani and Evrenosoglu in [6] used a combination of wavelet transform and SVM classifier to distinguish power system faults. The study used four binary SVM to classify the faults based on the energy of the wavelet-transform coefficients. Studies [7]–[12] used artificial neural networks (ANNs) for detection and classification of faults. In [7], the discrete-Fourier transform (DFT) was applied to the phase current and voltage signals, while the detection and classification of faults based on the discrete wavelet transform (DWT) and the ANN were studied by Silva *et al.* in [8]. Geethanjali and Priya in [9] used the detail and the approximation coefficients of DWT for the current signals as an input to ANN to classify the fault. In [10], the DWT, particle swarm optimization (PSO), and ANN were used to classify the power system faults. In [11], the spectral energy of the wavelet–transform coefficients are used after applying fourth-order Daubechies (db4) as the mother wavelet to detect the faults while ANN is used for fault classification. The combination of DFT and ANN was used to locate the faults in [12]. In [13], the KNN algorithm was used for fault classification and the fault occurrence was determined using the zero-sequence current signals. In [14], the combination of DWT and fuzzy logic was used for fault classification. Perez *et al.* [15] used an adaptive wavelet transform and the Bayesian linear discrimination analysis to classify faults. In [16], the DWT and DT classifier was used to classify the fault in microgrid (MG). In [17], a combination of self-organized maps with DT algorithms was used to locate transmission-lines (TLs) faults. The analysis presented in [5]–[17] was based on the assumption that all the data used for classification are labeled, hence the use of the SML approaches. Study [17] used two-stage classification by applying first an unsupervised learning technique to cluster the data followed by a supervised learning technique to perform the training and the classification. The main drawback of this approach is that the performance of the supervised learning classifier strongly depends on the outcome of the clustering process of the unsupervised learning technique, which strongly depends on the selection of the clustering parameters, and, hence, the classification accuracies may degrade when dealing with unlabeled data. In the presence of unlabeled data, the SML approaches and the combination of the unsupervised and supervised learning approaches fail to provide correct classification, and, therefore, there is a need for a SSML approach that can handle both labeled and unlabeled data.

The semisupervised learning techniques can be divided into three different methods. The first method uses self-training as in [18] and [19], in which each classifier learns on its own and relies only on its own predictions. The self-training method was used in nonintrusive load monitoring in [18] and in skin cancer diagnosis in [19]. The second method uses co-training as in [20]–[22], in which two basic classifiers are trained from the data source, which uses the most confident unlabeled data to be added to the labeled data in the learning process. The co-training method was applied in speech segmentation in [20], in human action recognition in [21], and in nonintrusive load monitoring in [22]. The third method is graph based as in [23]–[27], in which each sample diffuses its label information to its neighbors until a global stable state is obtained on the entire dataset. The graph-based method was utilized in image classification in [23]–[26], and in visual tracking in [27]. In this paper, the co-training algorithm is adopted to be applied to the fault detection and classification in both the transmission and the distribution systems with consideration of MG. In addition, the performance of the proposed co-training method is compared to that obtained using the other semisupervised methods (i.e., self-training and the graph-based methods).

B. Contribution

This paper introduces a SSML technique to classify the faults in TLs and MG using co-training of DT and KNN classifiers. Unlike previous works [5]–[17], in which only the SML approaches (e.g., SVM, ANN, KNN) were used with the analysis being limited to only labeled data, this study considers both labeled and unlabeled data using SSML. This study also considers various fault locations, types, impedances, and high-impedance fault (HIF). The study also demonstrates a systematic way to identify the most suitable mother wavelet and the number of wavelet decomposition levels, using the application of the harmony search algorithm (HSA).

C. Work Organization

This paper is organized as follows: Section II presents the SML versus the SSML; Section III describes the selection process of the wavelet(s) and the decomposition levels. The results are presented in Section IV and the conclusion is in Section V

II. SML Versus SSML

Classification techniques in general can be categorized according to the learning procedures into supervised and semisupervised learning techniques. The SML approaches usually learn only from the labeled data, while the SSML is characterized by an ability to learn from both labeled and unlabeled data.

A. Supervised Machine Learning

In SML, the labeled data in the dataset are usually partitioned into two subsets: one set for training and another set for testing. The SML approaches can be grouped as either eager learner (e.g., DT) or lazy learner (e.g., KNN). In case of DT, the inductive step involves the construction of a classification model (or a tree model) using the labeled training data, while in the deductive step, the tree model is applied to the labeled test data. The DT classifier is commonly known as an eager learner since it learns the labeled training data once they become available. On the other hand, in the case of KNN, only the training labeled data that resemble the attributes of the test examples are used, and, therefore, KNN is commonly described as a lazy learner.

1) DT Classifier: The DT model usually consists of three node types (i.e., leaf, internal, and root). The Gini index is used

Authorized licensed use limited to: Gheorghe Asachi Technical University of Ia¿i. Downloaded on January 26,2026 at 13:27:32 UTC from IEEE Xplore. Restrictions apply.

to estimate the node impurity; the index defines a better track in the tree to divide the attributes

$$\Phi(\beta) = 1 - \sum_{\alpha=0}^{C_a-1} [p(\alpha|\beta)]^2. \tag{1}$$

For a new record N_r, the class probability estimation $P_{c1}(\delta|N_r)$ can be calculated as follows:

$$P_{c1}(\delta|N_r) = \frac{\sum_{u=1}^{\varepsilon} \Theta(\delta_u, \delta) + 1}{\varepsilon + \varepsilon_t} \tag{2}$$

where $\Theta(\delta_u, \delta)$ is 1 if $\delta_u = \delta$ and zero otherwise. The class probability estimation for all classes is arranged in one vector as follows:

$$\Gamma_1(N_r) = [P_{c1}(\delta_1|N_r), \ldots\ldots, P_{c1}(\delta_{\varepsilon_t}|N_r)]. \tag{3}$$

2) KNN Classifier: The dataset D can be formed of a_t features with C cases and every case is allocated a label C_l, where the classes' number is m_c. The value of K determines the number of nearest neighbors to a new record. In the prediction process, the KNN classifier calculates the Euclidean distance (E) between the new record and the training dataset records with known labels

$$E_y = \sqrt{\sum_{z=1}^{a_t} (D_z^y - D'_z)^2} \qquad y = 1, \ldots, C \tag{4}$$

where the Euclidean distance between the new record D' and every record in dataset D^y at line y is E_y. The KNN records are found by defining the Kth smallest distance records with defining the label of every record. For multiclass problems, records which occur in the nearest neighbor follow a voting scheme (i.e., majority vote) to define the new record label

$$\tau_n = \underset{v=Cl_1,\ldots,Cl_{m_c}}{\arg\max} \sum_{z=1}^{K} \xi(\Omega_v = C_z), \xi \begin{cases} 1 & v = C_z \\ 0 & v \neq C_z \end{cases} \tag{5}$$

where $\xi(.)$ is an index which equal to 1 or 0 depending on Ω_v and C_z values. The class probability estimation for the new record $P_{c2}(\delta|N_r)$ can be calculated by using the following formula:

$$P_{c2}(\delta|N_r) = \frac{\sum_{m=1}^{K} \Psi(\Lambda_m, \Lambda) + 1}{K + m_c} \tag{6}$$

where $\Psi(\Lambda_m, \Lambda)$ is 1 if $\Lambda_m = \Lambda$ and zero otherwise. The class probabilities estimation for all classes are arranged in one vector as can be represented as

$$\Gamma_2(N_r) = [P_{c2}(\delta_1|N_r), \ldots\ldots, P_{c2}(\delta_{m_c}|N_r)]. \tag{7}$$

B. Semisupervised Machine Learning

In this study, the SSML using co-training of DT and KNN classifiers is applied to the detection and classification of faults. Algorithm 1, which is presented in the Appendix, explains in detail the co-training algorithm procedures. At first the dataset (attributes and classes) is divided into two groups: the first group λ contains the labeled attributes and the classes, while the second group θ only contains the unlabeled attributes. Each classifier generates a model M1 (in case of DT) and M2 (in case of KNN), respectively, by training on the same labeled data λ. A third group $\bar{g}$ is selected randomly from the unlabeled data and then the models M1 and M2 for DT and KNN classifiers are used to predict the classes for $\bar{g}$ by computing the class probability estimation $\Gamma_1(d)$ and $\Gamma_2(d)$ for each case d in group $\bar{g}$. The algorithm chooses from $\bar{g}$ (unlabeled data) the records that have the same prediction and the largest difference between $\Gamma_1(d)$ and $\Gamma_2(d)$ to update the labeled data so that each classifier is training the other classifier by its predicted classes. Both models M1 and M2 are then updated using the new labeled data and this process is repeated in each iteration until all the records in the unlabeled data group are used.

C. Classification Performance Evaluation

The k-fold cross validation is applied in this paper to estimate the classifier accuracy. The dataset is first partitioned into k groups of equal size; each group is then used for testing and the remaining groups are used for training. This process is repeated k trials. In every trial, the incorrectly predicted cases (Ξ) are recorded; classifier accuracy (Δ) of every trial k is then computed using the number of testing cases ρ_k

$$\Delta_K = \left(1 - \frac{\Xi}{\rho_K}\right). \tag{8}$$

The total accuracy of the classification model η is computed using

$$\eta = \frac{\sum_{j_o} \Delta_K(j_o)}{k}, \quad j_o = 1, 2 \ldots k. \tag{9}$$

III. Selection Process for the Wavelet(s) and the Decomposition Level(s)

In this paper, the features contained in the voltage and/or current waveforms are extracted with the DWT. The suitable wavelet function(s) are identified using the HSA after applying the DWT to the three-phase voltage and/or current signals. The energy of the wavelet coefficients is then computed and is subsequently used as the feature vector for class prediction after applying the co-training of the KNN and DT classifiers, as explained in Section III.

A. Discrete Wavelet Transform

The DWT offers a dyadic representation of the analyzed signal, which provides frequency subbands at a different resolution. This is a great advantage over the continuous wavelet transform (CWT) since only with DWT the MRA can be performed. The computational complexity of the DWT is only $O(n)$ where n is the data size, which is significantly less compared to that of the CWT and the undecimated wavelet transform (UWT), which are considered redundant transforms. The DWT contains an extensive library of wavelet basic functions, which makes this transform suitable for transient analysis. On the other hand, the Hilbert–Huang transform (HHT) has significant limitations in terms of generating undesired components in the low-frequency band, and a reduced ability to separate some low-energy components of the signal. Also, other transforms, such as the short-time

Authorized licensed use limited to: Gheorghe Asachi Technical University of Ia¿i. Downloaded on January 26,2026 at 13:27:32 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPUTATIONAL COMPLEXITY OF TRANSFORM ANALYSIS

Technique	UWT	DWT	HHT	CWT	WVD
Computational time	0.1954	0.0049	0.2410	0.2415	0.0807

Fourier transform and the Wigner–Ville distribution (WVD) suffer from the tradeoff between the time resolution and the frequency resolution. Table I lists the obtained computational times in seconds when the five transforms were run in MATLAB on an Intel Core i7, 2.9 GHz with 16-GB RAM personal computer (PC) and the results show the superiority of the DWT.

In DWT, the signals' features are extracted in both time and frequency domain using MRA. At each wavelet decomposition level *j*, the approximation wavelet coefficients Ca_j (low-frequency component) and the detail wavelet coefficients Cd_j (high-frequency component) are computed using

$$Ca_j(e) = \sum_{l_e} f_{l0}(l_e - 2e) Ca_{j-1}(l_e) \tag{10}$$

$$Cd_j(e) = \sum_{l_e} f_{l1}(l_e - 2e) Ca_{j-1}(l_e). \tag{11}$$

The energies of the approximations coefficients g_{aj} and the detail coefficients g_{dj} at the jth decomposition level can be calculated as follows:

$$g_{aj} = \sum_e |Ca_j(e)|^2 \quad \text{and} \quad g_{dj} = \sum_e |Cd_j(e)|^2. \tag{12}$$

The choice of the mother wavelet for the analysis strongly affects the values of the energy of the wavelet coefficients, which will have direct impact on the classification accuracy since it represents the feature vector. The literature has reported the use of different order of Daubechies wavelets (e.g., db1 in [16], db4 in [11], and db2 in [9]) without providing any justifications or even a systematic procedure. Moreover, the number of decomposition levels strongly affects the dimension of the feature vectors that contain the energies of the wavelet coefficients, another parameter that has never been clearly justified in the literature. Therefore, different wavelets can give different result, which triggers the need for a systematic approach that finds the optimal wavelet function(s) and the wavelet decomposition level combination. In this study, the HSA as an evolutionary optimization technique is utilized to systematically define the optimal wavelet(s) and the level(s) of decomposition for accurate fault classification. The HSA is used to explore the search space of the wavelet families. The algorithm searches for the suitable wavelet function for each level in each signal. Consequently, the computational complexity of the algorithm is *O* (number of signals [e.g., = 2] × number of decomposition level [e.g., = 5] × number of wavelet families [e.g., = 85]).

B. Feature Vectors Representation

The three-phase signals are sampled at 3.84 kHz (64 samples/cycle of power system frequency 60 Hz) and then the difference between each two successive cycles is calculated as

$$S_{ph}^d(j) = S_{ph}(j+n) - S_{ph}(j), \;\; j = 1, 2, \ldots, n. \tag{13}$$

The wavelet analysis is applied to the sequence S_{ph}^d and then the wavelet coefficients of the four decomposition levels (i.e., Cd_1, Cd_2, Cd_3, Cd_4, and Ca_4) and the energies of these coefficients are calculated using (10)–(12). The choice of the four decomposition levels is to guarantee that the 60-Hz component is centered in the approximation level. However, the number of wavelet decomposition levels that retains the features will be determined using HSA. The energies of the wavelet coefficients for the approximation (g_{a4}) and detail levels $(g_{d1} - g_{d4})$ for the six signals (three-phases voltage and three phases current) are arranged into one vector.

The energy of the wavelet coefficients are calculated for the following 11 different fault types: single line-to-ground faults (i.e., $a-g$, $b-g$, $c-g$); double line-to-ground faults (i.e., $ab-g$, $bc-g$, $ac-g$); double line faults (i.e., ab, bc, ac); three line faults (i.e., abc); and three line-to-ground faults (i.e., $abc-g$). As described previously in the DWT analysis steps, the calculated energies of all fault types (i.e., g_{ag}, g_{bg}, etc.) are arranged in one vector g_F. The vector of the energy of the wavelet coefficients is normalized. For example, in the case of single line-to-ground fault, the energy vector that corresponds to phase *a* to ground (g_{ag}) is normalized to Z_{ag} as

$$Z_{ag}(k) = \frac{g_{ag}(k) - \mu(g_{ag})}{\sigma(g_{ag})} \tag{14}$$

where $k = 1, 2, \ldots, 30$ represent the length of the energy vector g_{ag}. The vectors for the remaining faults are calculated in an identical manner as described in (14). The wavelet coefficient energies of the normal cases (no faults and under normal operating conditions) are computed by applying the DWT to the three-phase voltage and current signals followed by the normalization step to provide the vector Z_{gh}.

The Euclidean distance *EC* between the normal case vector Z_{gh} and the vector Z_g of every fault case in the g_F vector is computed as

$$EC_r = \sqrt{\sum_r \left[Z_{gt}(r) - Z_{gh}(r)\right]^2} \tag{15}$$

where $r = 1, 2, \ldots, 30$ represent the length of the energy vector Z_{gt}. The *EC* values for all fault types are arranged in one vector D_t where

$$D_t = [\,EC_{ag} \quad EC_{bg} \quad EC_{cg} \quad EC_{ab} \quad EC_{bc} \quad EC_{ac} \quad EC_{abg} \quad EC_{bcg} \quad EC_{acg} \quad EC_{abc} \quad EC_{abcg}\,]. \tag{16}$$

The variance σ_{Dt}^2 of the distance vector D_t is computed using

$$\sigma_{Dt}^2 = \frac{1}{m-1} \sum_q \left(D_t(q) - \mu(D_t)\right)^2 \tag{17}$$

where $q = 1, 2, \ldots, 30$ represent the length of the distance vector D_t. Since the variance $\sigma^2{}_{\text{Dis}}$ of the distance vector is affected by the choice of the wavelets and levels of decomposition utilized in the wavelet analysis, an optimization technique

Authorized licensed use limited to: Gheorghe Asachi Technical University of Ia¿i. Downloaded on January 26,2026 at 13:27:32 UTC from IEEE Xplore. Restrictions apply.

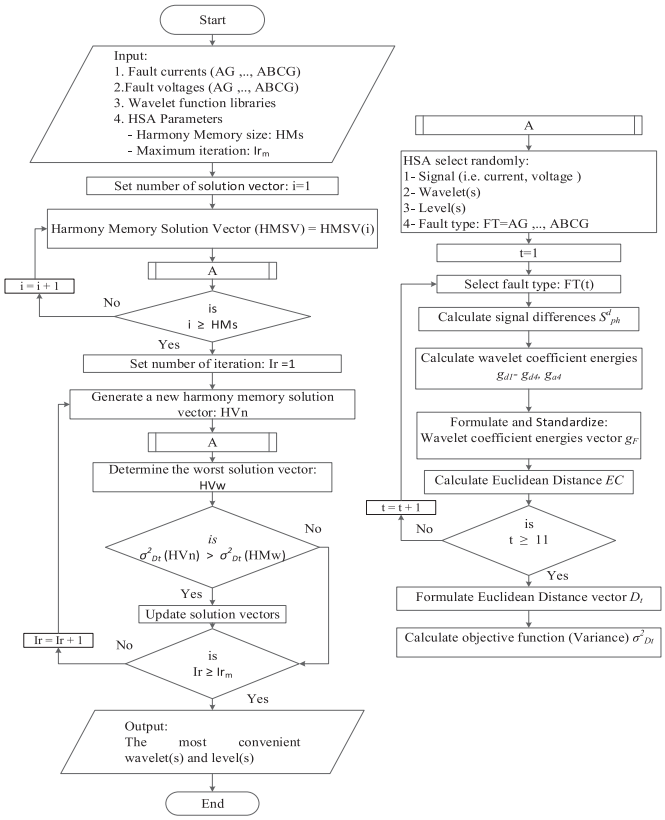

Fig. 1. Implementation of the HSA.

is needed. In this study, the HSA is applied to maximize the variance while searching for the optimal wavelet function(s) and the wavelet level(s).

C. Harmony Search Algorithm

The procedures of the HSA [26] can be outlined in the following steps: 1) define the objective function and decision variables, 2) initialize the harmony memory (HM) matrix, 3) generate a new solution vector, and 4) update the HM matrix. Fig. 1 shows the implementation procedures of HSA in detail.

IV. Results and Discussion

A. Test System Description

In order to evaluate the performance of the proposed approach, two different systems are used: test system 1 (TS-1) and test system 2 (TS-2).

The TS-1, which is outlined in [27], consists of a three-phase power source and a three-phase TL. Two three-phase transformers are used to feed two resistive loads (250, 100 MW) and a three-phase induction motor (100 MW). The test system is modeled in PSCAD/EMTDC software [28]. The current and voltage waveforms at Bus 1 (source bus) were recorded with sampling frequency 3.84 kHz (64 samples/cycle of network frequency 60 Hz). The sampling frequency chosen in this study is 3.84 kHz, which corresponds to 64 samples/cycle. This sampling rate represents the typical sampling rate used in most digital protective relays as reported by [29]. It is worth noting

TABLE II
Simulated Parameters of the Fault Cases of Test System-1

Fault Type	*abc*, *abcg*, *ab*, *bc*, *ac*, *ag*, *bg*, *cg*, *abg*, *bcg*, and *acg*
Fault Location	Ranges from 0 to 100 (% of the total line length)
Fault Resistance	Ranges from 0.01 to 200 Ω

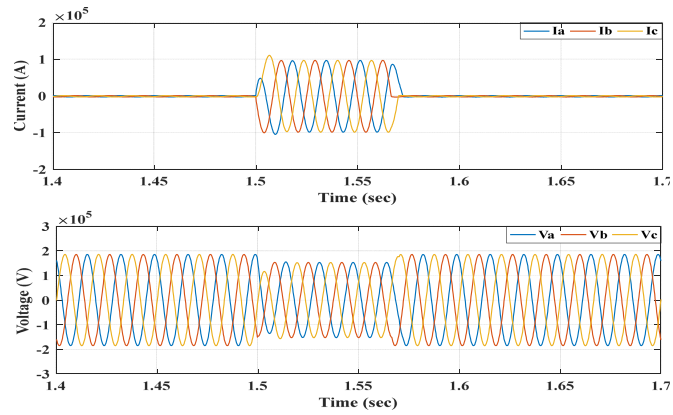

Fig. 2. Three-phase voltage and current at *abc* fault for TS-1.

TABLE III
Simulated Parameters of the Fault Cases of Test System-2

Fault Type	*abc*, *abcg*, *ab*, *bc*, *ac*, *ag*, *bg*, *cg*, *abg*, *bcg*, and *acg*
Fault Location	Ranges from 2 to 98 (% of the total line length)
Fault Resistance	Ranges from 0.1 to 10 Ω
Fault Duration	Ranges from 0.07 to 0.5 s
Fault Inception	Ranges from 0.1 to 0.9 s

that the proposed approach is flexible in terms of being adaptable to different sampling rates. For example, in the case of different sampling rates (e.g., 32 and 16 samples/cycle), the HSA search for the best combination of wavelet decomposition level(s) so to maximize the variances of the distance vector, as described in Section IV. The training and the testing datasets are generated following Table II, where 11 fault types are modeled with 23 fault resistance values ranging from 0.01 to 200 Ω at 12 different locations. This creates 3036 simulated records (i.e., 11 types × 12 locations × 23 resistances), which represent the total number of instances in the dataset. The three-phase signals in the case of *abc* fault are shown in Fig. 2.

The TS-2 is a real Brazilian 500-kV system and it consists of three TLs (i.e., 18.5, 151.5, and 325 km). The fault occurrence probability during the simulation is assumed the same for the three lines. The published dataset (UFPAFaults) of the fault events in [30] consists of 1000 cases. The waveforms were generated using EMTP-ATP [31] with a sampling frequency 40 kHz. In this study, the sampling frequency was downsampled to 3.84 kHz to be comparable with TS-1 data. All the values of the simulation parameters following Table III were drawn from a uniform probability density function. A complete description of the Brazilian system can be found in [32]. The three-phase voltage and current signals in the case of *bg* fault are in Fig. 3.

Authorized licensed use limited to: Gheorghe Asachi Technical University of Ia¿i. Downloaded on January 26,2026 at 13:27:32 UTC from IEEE Xplore. Restrictions apply.

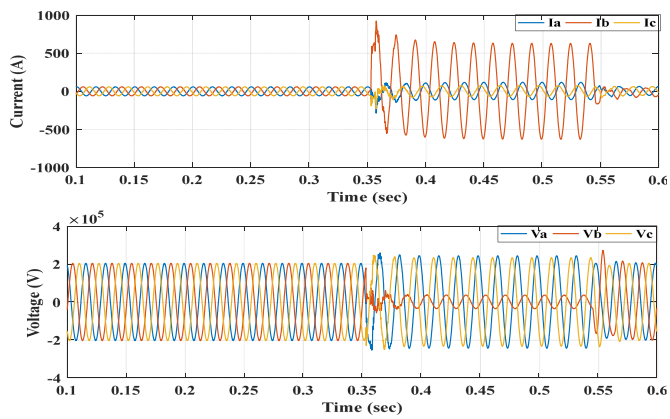

Fig. 3. Three-phase voltage and current at *bg* fault for TS-2.

TABLE IV
FITNESS FUNCTION VALUES

GA	PSO	HSA
Crossover rate: 0.8 Mutation rate: 0.01 Function: Roulette	Cognitive acceleration: 2 Social acceleration: 2 Min., Max. inertia: 0.4, 0.9	Consideration rate: 0.95 Pitch adjusting: 0.99 Bandwidth: 0.35
Fitness Value 3.495	3.480	4.470

TABLE V
BEST MOTHER WAVELET AND DECOMPOSITION LEVELS

Level (Voltage)	A_4	D_4	D_3	D_2	D_1
Wavelet	db11	db25	bior1.3	bior1.5	bior1.5
Level (Current)	A_4	D_4	D_3	D_2	D_1
Wavelet	sym4	db25	bior1.3	bior1.5	bior1.5

B. Systematic Procedures for Mother Wavelet Selection

The comparison between different optimization algorithms (e.g., genetic algorithm, HSA, PSO, and Tabu search) was performed in [33], and it was concluded that the HSA outperforms the other optimization algorithms. In this paper, in order to choose the best optimizer for the proposed method, the HSA, genetic algorithm (GA), and PSO algorithms were applied using the same number of population (i.e., 50) for each algorithm (e.g., individuals in GA, particles in PSO, and HM size in HSA). The remaining parameters of each algorithm were set as listed in Table IV. The fitness function tolerance value (i.e., 1×10^{-6}) was used as a termination criterion for all the algorithms. Table IV lists the fitness function values obtained using the three algorithms. The table shows HSA outperforming both GA and PSO by providing the maximum fitness function value. Table V lists the selected wavelet functions and the decomposition levels obtained when applying the HSA algorithm. It can be observed from the results listed in Table V that the HSA algorithm selects four decomposition levels (i.e., D_1, D_2, D_3, D_4, and A_4) for both the voltage and current signals and a specific wavelet function for each level. For example, the HSA algorithm chooses "db25" and the detail level 4 "D_4" for the voltage signal while "db11" was selected for the approximation level 4 "A_4."

C. Co-training and Performance Evaluation

The co-training of the DT and KNN classifiers is performed as a means of the SSML on the dataset containing the faulted and the nonfaulted cases, following Tables II and III. The percentage of the unlabeled data is commonly larger compared to the labeled data, and, therefore, in this study, for a total cases of 3036, the rate of labeled data is set to 2% (i.e., 60 cases) in the dataset and the unlabeled data represent nearly 98% (i.e., 2976 cases). The co-training algorithm starts by building the classifier model using the labeled data. The algorithm randomly chooses a subset without replacement 2% (i.e., 60 cases) from the unlabeled data. The new subset, which satisfied the updating conditions, is added to the labeled data and is then removed from the unlabeled data, as described in Section III. The training iterations terminate when all the unlabeled data cases are exhausted and cannot supply any further data to be added to the labeled dataset. For example, in each iteration, if all subset cases (i.e., 60 cases) were chosen from the unlabeled data (i.e., 2976 cases) and the updating conditions are satisfied, the iteration will then stop at 50 iterations (2975/60 $\approx$ 50). The classification accuracy is estimated using ten-fold cross validation for every iteration after the labeled data are updated. For example, in the first iteration, the initial labeled dataset (i.e., 60 cases) is used to build the classifier model and then the tenfold cross-validation method is applied to estimate the classifier accuracy by dividing the 60 cases into ten groups (i.e., six cases/group). The method starts by using one group for testing and the remaining nine groups for training. This step is repeated (ten times) until all groups are used for testing, as described in Section III. In the second iteration, the same procedures are repeated but the labeled dataset is updated by adding the unlabeled data until the iterations stop. The final accuracy of the co-training algorithm is considered as the classification accuracy at the last iteration after all unlabeled data were used and added to the labeled data.

D. Supervised, Unsupervised, and Semisupervised Accuracy Comparison

In order to assess the performance of the proposed approach, the results of the classification accuracies in the case of the SSML using co-training are compared to those obtained using: 1) SML as in [16]; 2) SSML using self-training as in [19], and 3) unsupervised machine learning as in [17]. In [16], the labeled data were only used to build the classifier model, while in [19], each classifier learned on its own and relies only on its own predictions. Also, in [17], all the unlabeled data were separated into groups according to their features and then the classification technique was applied. Equations (8) and (9) are used to compute the classification accuracies for each iteration in each trial. Figs. 4 and 5 depict the classification accuracies obtained in the case of DT and KNN when using the two test systems. It can be observed that as the iteration number increases, there is a significant improvement in the classification accuracies. The classification accuracies in case of the SML classifiers (i.e., no

Authorized licensed use limited to: Gheorghe Asachi Technical University of Ia¿i. Downloaded on January 26,2026 at 13:27:32 UTC from IEEE Xplore. Restrictions apply.

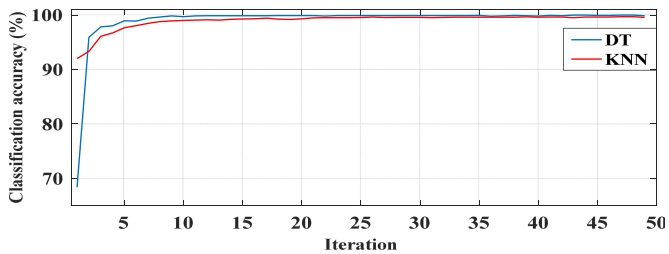

Fig. 4. DT and KNN classifier accuracy in case of 2-trial for TS-1.

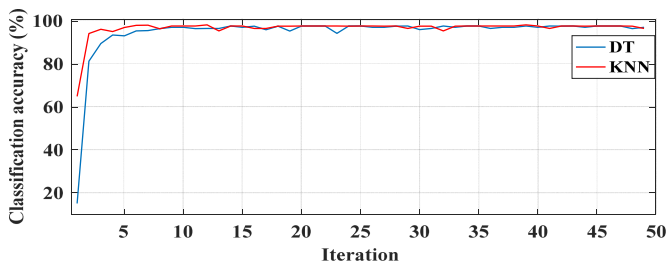

Fig. 5. DT and KNN classifier accuracy in case of 2-trial for TS-2.

TABLE VI
CLASSIFICATION ACCURACY OF TS-1 SYSTEM

Classifier	Overall accuracy (%)			
	SSML Co-training	SSML Self-Training	SML	Unsupervised
DT	99.88	96.84	78.15	96.57
KNN	99.69	95.01	96.16	

TABLE VII
CLASSIFICATION ACCURACY OF TS-2 SYSTEM

Classifier	Overall accuracy (%)			
	SSML Co-training	SSML Self-Training	SML	Unsupervised
DT	99.52	95.50	43.89	82.70
KNN	96.86	92.5	85.87	

co-training) and the SSML classifiers (i.e., using the proposed co-training approach) are computed and compared in Tables VI and VII. In case of TS-1, Table VI shows that the difference in the absolute value of the classifier accuracy between the proposed approach (semisupervised using co-training) and the supervised approaches are 3.53%, and 21.73% in the case of KNN and DT classifiers, respectively. It is clear that the use of the proposed semisupervised approach based on co-training significantly improves the classification accuracy compared to if only DT or KNN were used as in the SML classifier. On the other hand, according to Table VII, in the case of TS-2, the difference in the absolute value of the classifier accuracy between the proposed approach (semisupervised using co-training) and the supervised approaches are 10.99%, and 55.63% in the case of KNN and DT classifiers, respectively. These results confirm the observation made in Figs. 4 and 5 that in the case of the proposed SSML, the co-training of the two classifiers assists each other to better learn the pattern in the dataset, which results in better classification accuracies. The results also indicate that the proposed SSML algorithm was capable to handle and classify correctly all fault cases, as listed in Tables II and III, including the HIF cases, which are considered challenging in the power system protection.

TABLE VIII
CLASSIFICATION ACCURACY OF TS-1 SYSTEM WITH MEASUREMENT ERROR

Classifier	Overall Accuracy (%)			
	SSML Co-training	SSML Self-Training	SML	Unsupervised
DT	99.82	94.67	51.73	62.44
KNN	99.66	90.71	86.53	

TABLE IX
CLASSIFICATION ACCURACY OF TS-2 SYSTEM WITH MEASUREMENT ERROR

Classifier	Overall accuracy (%)			
	SSML Co-training	SSML Self-Training	SML	Unsupervised
DT	98.75	94.42	36.32	59.01
KNN	95.06	90.5	79.40	

E. Immunity Testing: Measurement Errors

The measurement errors were taken into account in this study by introducing a new dataset consisting of 120 cases for each test system and which incorporates measurement errors within the range $\pm 3\%$ following the IEEE/ANSI Standard. Tables VIII and IX list the obtained classification accuracies in the two test systems. The tables show again the proposed semisupervised learning algorithm outperforming the other approaches, which confirms the immunity of the proposed approach to the measurement errors.

F. MG Testbed System

The MG protection represents the main challenge facing its operation, in particular when considering the integration of distributed energy resources (DERs). In this paper, the performance of the proposed co-training approach was also examined on a MG system. The consortium for electric reliability technology solutions (CERTS), which is outlined in [34] and is shown in Fig. 6, was used as a case study of a MG system in this paper. The CERTS MG can be operated in two modes (i.e., grid-connected (GC) or in nongrid-connected (NGC) islanded mode). The CERTS MG is considered as a part of a distribution system, which is supplied from the secondary (low-voltage side) of three-phase distribution transformers rated 13.8/0.48 kV in the case of the GC mode or is supplied from the DERs in the case of the islanded mode. As shown in Fig. 6, the CERTS MG distribution system consists of three DERs, two solar photovoltaic sources (DER-PV1, DER-PV2), and one battery energy storage source (DER-Bt.S). Four loads (L3, L4, L5, and L6) are distributed along the system.

Authorized licensed use limited to: Gheorghe Asachi Technical University of Ia¿i. Downloaded on January 26,2026 at 13:27:32 UTC from IEEE Xplore. Restrictions apply.

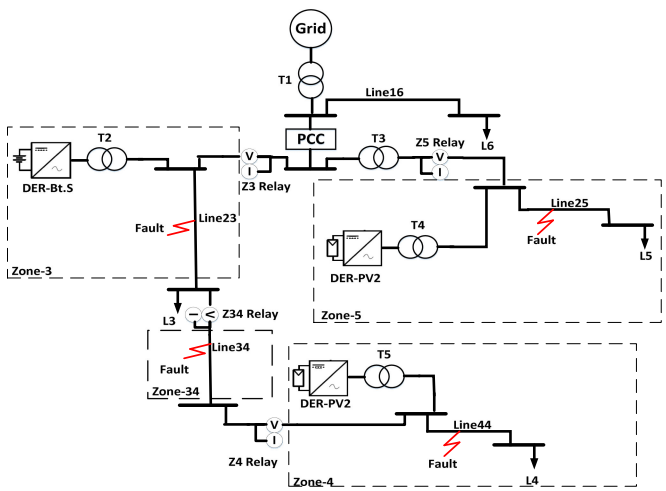

Fig. 6. CERTS MG system structure.

TABLE X
SIMULATED PARAMETERS OF THE FAULT CASES OF MG

Fault Type	*abc*, *abcg*, *ab*, *bc*, *ac*, *ag*, *bg*, *cg*, *abg*, *bcg*, and *acg*
Fault Location	Z5, Z4, Z3, and Z34
Fault Resistance	Ranges from 0.1 to 10 Ω
Mode of operation	Islanded and GC

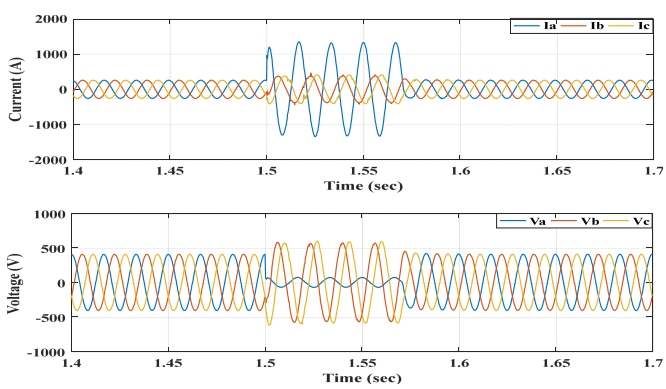

Fig. 7. Three-phase voltage and current at *ag* fault for MG.

The CERTS MG is modeled using the PSCAD/EMTDC software. The voltage/current signals are sampled at 64 samples/cycle of grid frequency 60 Hz (i.e., sampling frequency 3.84 kHz). The sampled signals are recorded at each protection relay. The training and the testing datasets are generated following Table X, where 11 fault types are modeled with 12 fault resistance values at four different locations and two operating modes. This creates 1056 simulated records (i.e., 11 types × 4 locations × 12 resistances × 2 modes), which represent the total number of instances in the dataset. The three-phase voltage and current signals in the case of *ag* fault at Z5 relay with NGC mode are shown in Fig. 7.

Table XI lists the selected wavelet functions and the decomposition levels obtained when applying the HSA algorithm in the MG system. It can be observed from the results listed in Table XI that the HSA algorithm selects four decomposition levels (i.e., D_1, D_2, D_3, D_4, and A_4) for the current signal, and three decomposition levels (i.e., D_1, D_2, and D_3) for the voltage signal. It has been observed that the proposed algorithm (HSA-based) was able to identify the most suitable wavelet functions and the optimal number of wavelet decomposition levels, which show the adaptability/flexibility of the proposed algorithm. Specifically, as listed in Table XI, the features in the voltage signal are contained within the detail levels $(D_1 - D_3)$, while in the case of the current signal, the HSA has identified all four levels including the approximation and the details. Table XII lists the obtained classification accuracies in the MG system. Again, Table XII shows the proposed semisupervised learning algorithm outperforming the other approach (i.e., supervised approaches), which confirms the strength of the proposed approach even when applied to a different system topology.

TABLE XI
BEST MOTHER WAVELET AND DECOMPOSITION LEVELS IN MG

Level (Voltage)	A_4	D_4	D_3	D_2	D_1
Wavelet	–	–	sym9	bior1.5	db41
Level (Current)	A_4	D_4	D_3	D_2	D_1
Wavelet	db8	db4	Bior3.1	Bior3.5	bior1.3

TABLE XII
CLASSIFICATION ACCURACY OF MG SYSTEM

Classifier	Overall Accuracy (%)			
	SSML Co-training	SSML Self-Training	SML	Unsupervised
DT	97.81	91.91	35.25	62.44
KNN	96.70	86.67	74.78	

The main objective of the feature selection methods is to identify the optimal feature subset in order to achieve the best classification performance. The wrapper approach [35] is the most commonly used approach for the feature selection, which mainly relies on the search-based techniques, among which the PSO was used in [36], and the GA was used in [37]. In this paper, the proposed HSA as an optimization technique is used to search for the most suitable wavelet functions and the optimal number of wavelet decomposition levels. Hence, the energy of the wavelet coefficients, which is computed using the identified wavelet functions and wavelet decomposition levels, is used as the input feature to the classifier. Tables V and XI reveal the capability of the proposed algorithm (HSA-based) to extract the feature by identifying the optimal decomposition levels holding the salient features for each studied system (transmission system and MG system).

G. Comparison With Graph-Based Semisupervised Method (GBSSM)

The performance of the proposed co-training approach was also compared with the GBSSM. The procedures of GBSSM [23] can be outlined in the following steps.

Authorized licensed use limited to: Gheorghe Asachi Technical University of Ia¿i. Downloaded on January 26,2026 at 13:27:32 UTC from IEEE Xplore. Restrictions apply.

TABLE XIII
CLASSIFICATION ACCURACY OF GBSSM TECHNIQUE

Testbed System	GBSSM Classification Accuracy (%)		
	Minimum	Average	Maximum
TS-1	18.15	66.24	94.56
TS-2	17.11	61.91	79.57
MG	2.27	29.29	79.48

1) Initialize the dataset matrix G using the labeled data where the G matrix consists of a_t attributes with R cases and each record is assigned a class label C with a total number of classes n_c

$$G = \begin{bmatrix} {G_1}^1 & .. & {G_{a_t}}^1 \\ .. & .. & .. \\ {G_1}^R & .. & {G_{a_t}}^R \end{bmatrix}_{R \times a_t}. \tag{18}$$

2) Initialize the label matrix Y using

$$Y(i,j) = \begin{cases} 1, & \mathrm{C}_i = j \quad \mathrm{j} \in \{1, ..., n_c\} \\ 0, & \text{otherwise} \end{cases},$$

$$Y = \begin{bmatrix} 1 & .. & 0 \\ .. & .. & .. \\ 0 & .. & 1 \end{bmatrix}_{R \times n_c}. \tag{19}$$

3) Construct the weight/adjacency matrix W according to (20), where σ is the bandwidth parameter

$$W(i,j) = \begin{cases} \exp(-\|G_i - G_j\|^2/2\sigma^2), & i \neq j \\ 0, & i = j \end{cases}. \tag{20}$$

4) Construct the diffusion kernel matrix S according to

$$S = D^{-1/2}\, W\, D^{-1/2} \text{ and } D = \begin{cases} \sum_{\mathrm{j}} W(i,j) & i = j \\ 0 & i \neq j \end{cases}. \tag{21}$$

5) Construct the classification solution matrix F according to (22), where $0 < \alpha < 1$

$$F = (I - \alpha S)^{-1} Y,\ F = \begin{bmatrix} {f_1}^1 & .. & {f_{n_c}}^1 \\ .. & .. & .. \\ {f_1}^R & .. & {f_{n_c}}^R \end{bmatrix}_{R \times n_c}. \tag{22}$$

6) Update the matrices G, W, S, and F using the unlabeled dataset. In addition, update the label matrix Y with new label of the unlabeled data according to the classification solution matrix F using the following formula:

$$y_i = \arg\max_{j \leq n_c} f(i,j), \quad j \in \{1, ..., n_c\}. \tag{23}$$

Table XIII lists the obtained classification accuracies in the TS-1, TS-2, and MG systems using the GBSSM technique with the parameter $\sigma = 1$ as suggested in [23], and the parameter α was varied from 0 to 1 in 0.1 step. It can be observed from

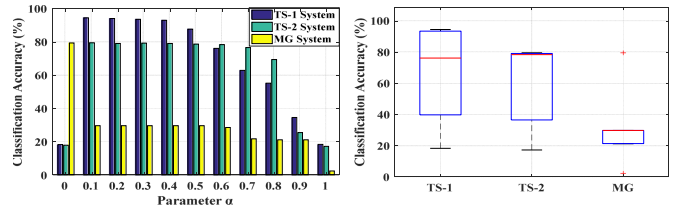

Fig. 8. GBSSM classifier accuracy in case of different value of α.

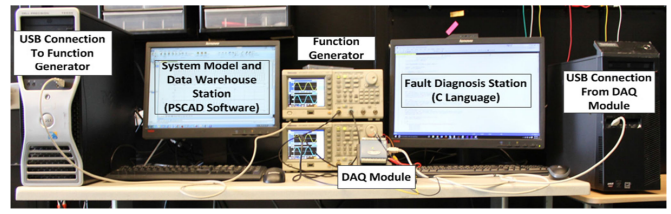

Fig. 9. Real-time experimental setup.

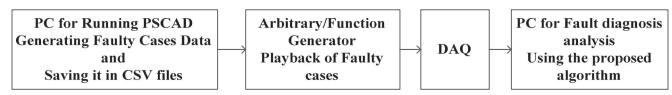

Fig. 10. Schematic block diagram of the experimental setup

Fig. 8 that the performance of the GBSSM is strongly affected by the choice of the parameters σ and α, which is considered a major drawback of the GBSSM. On other hand, the results listed in Table XIII show again the proposed co-training technique outperforming the GBSSM technique, which demonstrates the effectiveness of the proposed co-training approach.

H. Experimental Work

In order to verify the effectiveness of the proposed approach, the experimental setup shown in Fig. 9 was used. The following is the hardware description: 1) arbitrary/function generator: Tektronix AFG3022C, 25-MHz sine waveforms and 14-bits arbitrary waveform. The AGF are typically used with the supporting software, which allows importing any signal and playing it back in real time, 2) data acquisition module (DAQ): Omega 1608FS-PLUS, eight channel, 16-bit multifunction universal serial bus (USB). It can sample up to eight channels at a maximum speed of 50 kHz per channel or four channels at 100 kHz, while delivering 16-bit accuracy, and 3) PC: Intel(R) CPU E5520 at 2.27 GHz, Dual core, 14-GB RAM. Fig. 10 illustrates the schematic block diagram showing the procedures of the experimental execution.

The PSCAD software was employed on the PC in order to generate new faulty cases and then the collected data were stored in as a comma separated values (CSV) format. Through the USB connection, the function generator reads the CSV file for the voltage and current, and then plays it back in real time. The output signal of the function generator is supplied to the DAQ module as an input. The input signal to the DAQ module is then sampled by the DAQ at a sampling rate of 64 samples/cycle. The DAQ module then transfers the sampled data to the PC via USB. The proposed algorithm for the fault diagnosis was

Authorized licensed use limited to: Gheorghe Asachi Technical University of Ia¿i. Downloaded on January 26,2026 at 13:27:32 UTC from IEEE Xplore. Restrictions apply.

TABLE XIV
EXPERIENTIAL RESULTS IN CASE OF THE PROPOSED APPROACH

Parameters		
Fault Type: *ag*, *bg*, *cg*, *ab*, *bc*, *ac*, *abg*, *bcg*, *acg*, *abc*, and *abcg*		
Fault Location: 0, 50, 100 km, and Fault Resistance: 0.01, 10, 100 Ω		
Classifier	Misclassified cases	Accuracy (%)
DT	0 out of 20	100
KNN	1 out of 20	95

implemented and written on the PC using the C-language, which can be easily implemented on a microcontroller. A new set of faulty cases were generated at various operating conditions is listed in Table XIV. The total number of cases is 20 and each of these cases were mimicked by the function generator and transferred to the PC after the DAQ module had completed the sampling process. It is evident in the table that follows that the proposed algorithm is capable of successfully detecting and classifying the faults for each of the 20 cases. In the case of DT, the proposed algorithm was able to detect and classify 20 cases correctly. While the KNN was able to detect and classify 19 cases correctly. The results also show that DT outperforming the KNN classifier and providing 100% classification accuracy, which confirms the classification accuracy listed in Table VI, and, hence, confirming the effectiveness of the proposed approach in real-time implementation.

V. CONCLUSION

This paper presented a SSML approach using co-training of DT and KNN classifiers applied to fault classification in power systems. In this paper, the co-training of DT and KNN classifiers was used to establish the classification model based on the labeled and unlabeled data for automating the fault classification process. The DWT is applied to extract the prominent features in the current and voltage waveforms after identifying the optimal mother wavelets and the levels of decomposition using the HSA. Unlike the SML approaches, the SSML using co-training is able to handle both the labeled and unlabeled data instead of only the labeled data. The energy of the wavelet coefficients computed using DWT represents the feature vector to which the HSA is applied, hence identifying the most suitable mother wavelet and the wavelet decomposition levels for fault classification. The proposed semisupervised approach using the co-training algorithm is tested on two different test systems including 11 fault types and the classification accuracy was evaluated using tenfold cross validation. The results have shown that the proposed semisupervised approach was able to achieve significant improvement in the classification accuracies compared to the SML approach in the case of using unlabeled data to train and update the classifier model. The obtained differences in the absolute value between the accuracy of the proposed approach (co-training as SSML) and the SML were found to be 3.53%, and 21.73% in the case of KNN and DT, respectively.

APPENDIX

Algorithm 1: Co-training Algorithm.

1: **Start, Inputs:** DT classifier A_1, KNN classifier A_2, and dataset
2: Divided the dataset into: λ (labeled data) and θ (unlabeled data)
3: Assign $\pounds_x = \lambda$, $\pounds_y = \lambda$
4: Induce: M_1 with training A_1 on $\pounds_x$ and M_2 with training A_2 on $\pounds_y$
5: Generate a subset $\bar{g}$ by choosing randomly $\bar{G}$ data from θ
6: **While** (size $(\theta) > \bar{G}$) **Do**
7: Predict the class for each data d in $\bar{g}$ using M_1 model and compute $\Gamma_1(d)$
8: Predict the class for each data d in $\bar{g}$ using M_2 model and compute $\Gamma_2(d)$
9: Check for the updating conditions where
$-\mathfrak{l}_x$ choose from $\bar{g}$ with highest differences of class probability:

$$d = \arg\max_d(\Gamma_2(d) - \Gamma_1(d)) \tag{A.1}$$

$-\mathfrak{l}_y$ choose from $\bar{g}$ with highest differences of class probability:

$$d = \arg\max_d(\Gamma_1(d) - \Gamma_2(d)) \tag{A.2}$$

10: Update $\pounds_x = \pounds_x + \mathfrak{l}_x$, $\pounds_y = \pounds_y + \mathfrak{l}_y$ and remove $\mathfrak{l}_x$, $\mathfrak{l}_y$ from $\bar{g}$
11: Select randomly a new $\bar{G}$ data from θ to refill $\bar{g}$
12: Update M_1 by using A_1 to train on $\pounds_x$ and M_2 by using A_2 to train on $\pounds_y$
13: **End While**
14: **Output:** The final model of KNN and DT classifier
15: **End**

REFERENCES

[1] J. Morais, Y. Pires, C. Cardoso, and A. Klautau, "A framework for evaluating automatic classification of underlying causes of disturbances and its application to short-circuit faults," *IEEE Trans. Power Del.*, vol. 25, no. 4, pp. 2083–2094, Oct. 2010, doi: 10.1109/TPWRD.2010.2052932.

[2] A. Rahmati and R. Adhami, "A fault detection and classification technique based on sequential components," *IEEE Trans. Ind. Appl.*, vol. 50, no. 6, pp. 4202–4209, Dec. 2014, doi: 10.1109/TIA.2014.2313652.

[3] Q. Alsafasfeh, I. Abdel-Qader, and A. Harb, "Fault classification and localization in power systems using fault signatures and principal components analysis," *Energy Power Eng.*, vol. 4, no. 6, pp. 506–522, Nov. 2012, doi: 10.4236/epe.2012.46064.

[4] M. J. B. Reddy, D. V. Rajesh, and D. K. Mohanta, "Robust transmission line fault classification using wavelet multi-resolution analysis," *Comput. Electr. Eng.*, vol. 39, no. 4, pp. 1219–1247, May 2013, doi: 10.1016/j.compeleceng.2013.02.013.

[5] P. Gopakumar, M. J. B. Reddy, and D. K. Mohanta, "Adaptive fault identification and classification methodology for smart power grids using synchronous phasor angle measurements," *IET Gener. Transm. Distrib.*, vol. 9, no. 2, pp. 133–145, Jan. 2015, doi: 10.1049/iet-gtd.2014.0024.

[6] H. Livani and C.Y. Evrenosoglu, "A fault classification method in power systems using DWT and SVM classifier," in *Proc. IEEE PES Transm. Distrib. Conf. Expo.*, May 2012, pp. 1–5, doi: 10.1109/TDC.2012.6281686.

[7] M. Ben Hessine, H. Jouini, and S. Chebbi, "Fault detection and classification approaches in transmission lines using artificial neural networks," in *Proc. 17th IEEE Med. Electrotech. Conf.*, Apr. 2014, pp. 515–519, doi: 10.1109/MELCON.2014.6820588.

Authorized licensed use limited to: Gheorghe Asachi Technical University of Ia¿i. Downloaded on January 26,2026 at 13:27:32 UTC from IEEE Xplore. Restrictions apply.

[8] K. M. Silva, B. A. Souza, and N. S. D. Brito, "Fault detection and classification in transmission lines based on wavelet transform and ANN," *IEEE Trans. Power Del.*, vol. 21, no. 4, pp. 2058–2063, Oct. 2006, doi: 10.1109/TPWRD.2006.876659.

[9] M. Geethanjali and K.S. Priya, "Combined wavelet transfoms and neural network (WNN) based fault detection and classification in transmission lines," in *Proc. Int. Conf. Control, Autom., Commun. Energy Conserv.*, Jun. 2009, pp. 1–7.

[10] J. Upendar, C. P. Gupta, G. K. Singh, and G. Ramakrishna, "PSO and ANN-based fault classification for protective relaying," *IET Gener. Transm. Distrib.*, vol. 4, no. 10, pp. 1197–1212, Oct. 2010, doi: 10.1049/iet-gtd.2009.0488.

[11] J. Chen and R. K. Aggarwal, "A new approach to EHV transmission line fault classification and fault detection based on the wavelet transform and artificial intelligence," in *Proc. IEEE Power Eng. Soc. Gen. Meet.*, San Diego, CA, USA, Jul. 2012, pp. 1–8, doi: 10.1109/PESGM.2012.6344762.

[12] A. A. Dutta, A. N. Kadu, and M. M. Rao, "Intelligent control for locating fault in transmission lines," *Int. J. Instrum., Contr. Autom.*, vol. 1, no. 2, pp. 64–70, Jul. 2011.

[13] A. Adav and A. Swetapadma, "Fault analysis in three phase transmission lines using k-nearest neighbor algorithm," in *Proc. Int. Conf. Adv. Electron., Comput. Commun.*, Oct. 2014, pp. 1—5, doi: 10.1109/ICAECC.2014.7002474.

[14] M. Jamil, R. Singh, and S. K. Sharma, "Fault identification in electrical power distribution system using combined discrete wavelet transform and fuzzy logic," *J. Electr. Syst. Inf. Technol.*, vol. 2, no. 2, pp. 257–267, Sep. 2015, doi: 10.1016/j.jesit.2015.03.015.

[15] F. E. Peérez, E. Orduña, and G. Guidi, "Adaptive wavelets applied to fault classification on transmission lines," *IET Gener. Transm. Distrib.*, vol. 5, no. 7, pp. 694–702, Jul. 2011, doi: 10.1049/iet-gtd.2010.0615.

[16] D. P. Mishra, S. R. Samantaray, and G. Joos, "A combined wavelet and data mining based intelligent protection scheme for microgrid," *IEEE Trans. Smart Grid*, vol. 7, no. 5, pp. 2295–2304, Sep. 2016, doi: 10.1109/TSG.2015.2487501.

[17] C. Pothisarn and A. Ngaopitakkul, "The combination of discrete wavelet transform and self-organizing map for identification of fault location on transmission line," in *Proc. Int. Multiconf. Eng. Comput. Scientists*, Mar. 2012, vol. 2, no. 1, pp. 1083–1086.

[18] A. Iwayemi and C. Zhou, "SARAA: Semi-supervised learning for automated residential appliance annotation," *IEEE Trans. Smart Grid*, vol. 8, no. 2, pp. 779–786, Mar. 2017, doi: 10.1109/TSG.2015.2498642.

[19] A. Masood, A. Al- Jumaily, and K. Anam, "Self-supervised learning model for skin cancer diagnosis," in *Proc. Int. IEEE/EMBS Conf. Neural Eng.*, Apr. 2015, pp. 1012–1015, doi: 10.1109/NER.2015.7146798.

[20] U. Guz, S. Cuendet, D. Hakkani-Tur, and G. Tur, "Multi-view semi-supervised learning for dialog act segmentation of speech," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 18, no. 2, pp. 320–329, Feb. 2010, doi: 10.1109/TASL.2009.2028371.

[21] C. Liu and P. Yuen, "A boosted co-training algorithm for human action recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 21, no. 9, pp. 1203–1213, Sep. 2011, doi: 10.1109/TCSVT.2011.2130270.

[22] J. M. Gillis and W. G. Morsi, "Non-intrusive load monitoring using semi supervised machine learning and wavelet design," *IEEE Trans. Smart Grid*, vol. PP, no. 99, pp. 1–8, Mar. 2016, doi: 10.1109/TSG.2016.2532885.

[23] B.-B. Liu and Z.-M. Lu, "Image colourisation using graph-based semisupervised learning," *IET Image Process.*, vol. 3, no. 3, pp. 115–120, 2009, doi: 10.1049/iet-ipr.2008.0112.

[24] Z. Zhang, M. Zhao, and T. W. S. Chow, "Graph-based constrained semisupervised learning framework via label propagation over adaptive neighborhood," *IEEE Trans. Knowl. Data Eng.*, vol. 27, no. 9, pp. 2362–2376, Sep. 2015, doi: 10.1109/TKDE.2013.182.

[25] W. Hu, J. Gao, J. Xing, C. Zhang, and S. Maybank, "Semi-supervised tensor-based graph embedding learning and its application to visual discriminant tracking," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 1, pp. 172–188, Jan. 2017, doi: 10.1109/TPAMI.2016.2539944.

[26] Z. W. Geem, "Harmony search algorithm for solving sudoku," in *Proc. Int. Conf. Knowl.-Based Intell. Inf. Eng. Syst.*, 2007, vol. 4692, pp. 371–378, doi: 10.1007/978-3-540-74819-9_46.

[27] J. A. Jiang *et al.*, "Hybrid framework for fault detection, classification, and location Part I: Concept, structure, and methodology," *IEEE Trans. Power Del.*, vol. 26, no. 3, pp. 1988–1997, Jul. 2011, doi: 10.1109/PESMG.2013.6672597.

[28] *PSCAD/EMTDC User's Guide*, HVDC Research Centre, Winnipeg, MB, Canada, Apr. 2005.

[29] M. Kezunovic, J. Ren, and S. Lotfifard, *Design, Modeling and Evaluation of Protective Relays for Power Systems*, 1st ed. Cham, Switzerland, Springer Int. Publishing, 2016, doi: 10.1007/978-3-319-20919-7.

[30] *Signal Processing Laboratory—Federal Univ. Para, UFPAFaults*. 2009 [Online]. Available: http://www.laps.ufpa.br/freedatasets/UfpaFaults

[31] EMTP. Alternative Transient Program (ATP) Rule Book, Canadian/American EMTP User's Group, 1995.

[32] Y. Pires *et al.*, "A framework for evaluating data mining techniques applied to power quality," in *Proc. Brazilian Conf. Neural Netw.*, 2005, pp. 193–198, doi: 10.21528/CBRN2005-193.

[33] K. S. Lee and Z. W. Geem, "A new metaheuristic algorithm for continuous engineering optimization: Harmony search theory and practice," *Comput. Methods Appl. Mech. Eng.*, vol. 194, nos. 36–38, pp. 3902–3933, Sep. 2004, doi: 10.1016/j.cma.2004.09.007.

[34] R. H. Lasseter *et al.*, "CERTS microgrid laboratory test bed," *IEEE Trans. Power Del.*, vol. 26, no. 1, pp. 325–332, Jan. 2011, doi: 10.1109/TPWRD.2010.2051819.

[35] L. Y. Huan Liu, "Toward integrating feature selection algorithms for classification and clustering," *IEEE Trans. Knowl. Data Eng.*, vol. 17, no. 4, pp. 491–502, Apr. 2005, doi: 10.1109/TKDE.2005.66.

[36] Y. Zhang, S. Wang, P. Phillips, and G. Ji, "Binary PSO with mutation operator for feature selection using decision tree applied to spam detection," *Knowl. Based Syst.*, vol. 64, pp. 22–31, Jul. 2014, doi: 10.1016/j.knosys.2014.03.015.

[37] Y. Z. Jianjiang Lu and T. Zhao, "Feature selection based-on genetic algorithm for image annotation," *Knowl. Based Syst.*, vol. 21, no. 8, pp. 887–891, 2008, doi: org/10.1016/j.knosys.2008.03.051.

Tamer S. Abdelgayed (S'15) was born in Cairo, Egypt, in 1981. He received the B.Sc. (Eng.) and M.Sc. (Eng.) degrees in electrical engineering from Helwan University, Cairo, in 2003 and 2007, respectively. He is currently working toward the Ph.D. degree in electrical engineering at the Faculty of Engineering and Applied Science, University of Ontario Institute of Technology, Oshawa, ON, Canada.

His research interests include power system protection, data analytics, and automation in smart grid.

Mr. Abdelgayed is a Registered Engineer in Training with the Association of Professional Engineers in Ontario.

Walid G. Morsi (S'07–M'09–SM'16) was born in Ismailia, Egypt, in 1975. He received the B.Sc. (Eng.) and M.Sc. (Eng.) degrees in electrical engineering from Suez Canal University, Ismailia, in 1998 and 2002, respectively, and the Ph.D. degree in electrical engineering from Dalhousie University, Halifax, NS, Canada, in 2009.

He was a Killam Memorial Predoctoral Scholar with Dalhousie University, and then he was an Assistant Professor with the Department of Electrical and Computer Engineering, University of New Brunswick, Fredericton, NB, Canada. He is currently an Associate Professor with the Department of Electrical Computer and Software Engineering, Faculty of Engineering and Applied Science, University of Ontario Institute of Technology, Oshawa, ON, Canada. His research interests include smart grid, power quality/disturbance data analytics, transportation electrification, energy monitoring, management, and automation of electric power systems.

Dr. Morsi is a Registered Professional Engineer with the Association of Professional Engineers in Ontario.

Tarlochan S. Sidhu (M'90–SM'94–F'04) received the Ph.D. degree in electrical engineering from the University of Saskatchewan, Saskatoon, SK, Canada, in 1989.

He was a Professor and Chair with the Electrical and Computer Engineering Department, University of Western Ontario, London, ON, Canada. He is currently a Professor and Dean of the Faculty of Engineering and Applied Science, University of Ontario Institute of Technology, Oshawa, ON, Canada. He also has held the NSERC/Hydro One Senior Industrial Research Chair in Power Systems Engineering. His research interests include power system protection, monitoring, control, and automation.

Authorized licensed use limited to: Gheorghe Asachi Technical University of Ia¿i. Downloaded on January 26,2026 at 13:27:32 UTC from IEEE Xplore. Restrictions apply.